

Read me first — sim re-entry (540, July 2026)

Last synced: 2026-07-27 — ρ assortativity, archive manifest, Pass B ablation spec.

OPORD: `../3-Master_Plan/re_entry/model_OPORD.md`

If you feel lost: read `../3-Master_Plan/re_entry/00_READ_ME_FIRST.md` (docs 01–03) before opening sim code.

Revolution vs evolution

Do	Don't
Import <code>tier1_pool_assignment.py</code> , <code>tier1_generative_edu.py</code> , <code>tier1_sim_config.py</code>	Extend archived notebooks (<code>538D</code> , <code>538</code> , <code>537</code> , <code>535</code>)
Thin scripts / <code>540_*</code> notebook	Copy-paste 538D cells
Answer Alex-sized questions (knockouts, ablations)	Rivanna sweeps, bin-for-bin hero match as v1 gate

Archived notebooks in `sports/archive/` are **SCOUT-era lab reference** — read only, do not edit.

Charles’s three-step pipeline

- 1. **Assign** — build rosters (ρ or sort-and-chop)
- 2. **Score** — $S_i = A_i - \lambda \cdot L_C$ (Alex v1: $(B-D)=-L_C$ in selection)
- 3. **Draft** — top K by S_i

Knockouts (separate experiments):

Pass	What toggles	Status
A	$\lambda=0$ vs congestion in S_i	Done → <code>alex_side_by_side_v0/</code>
B	ρ low / ρ high / sort-and-chop; fix selection	<code>540_rho_ablation_bundle.py</code>

Three-step assignment (step 1 only)

- 0. Draw latent abilities A_i
 - 1. Draw team targets T_j (e.g. Uniform on $[-0.5, 0.5]$)
 - 2. Sequential soft match with roster caps:

$$\pi_{ij} \propto \exp \left(-\rho \cdot \frac{(A_i - T_j)^2}{2\sigma^2} \right) \times (n_j + k)^\alpha$$

Optional preferential attachment: `USE_PREFERENTIAL_ATTACHMENT = True` → $\alpha > 0$; default **False**.

ρ	Meaning
ρ = 0	Uniform among teams with roster room (max mixing)
ρ = 1	Moderate assortativity (≡ legacy $\tau = \sigma \approx 0.65$)
ρ high	Sharper match to nearest T_j
sort-and-chop	Separate code path — 537-style global sort + equal slices; max assortativity benchmark , not $\rho \rightarrow \infty$

σ (`ASSIGNMENT_SIGMA`, default **0.65**) is a fixed scale — not the user-facing knob.

Legacy docs use τ (**temperature**) with **opposite** intuition (small τ = assortative). **540+ uses ρ only** in prose and exports.

Hero estimand lock

Field	Value
Outcome	<code>Y_draft</code> , mean draft rate per bin
X (hero)	<code>poolq_loo</code>
Bins	16 quantile
Winsor	(0.01, 0.99)
min_minutes	20
Seasons	2011–2021

Slug: `empirical_ppm_poolq_loo_16quantile_winsor0199_min20_2011`

Active files

Role	Path
Empirical	530_sports_pipeline.ipynb , sports_pipeline/
Sim engines	tier1_pool_assignment.py , tier1_generative_eda.py , tier1_sim_config.py
Pass A bundle	scripts/hero_model_reset_bundle.py → alex_side_by_side_v0/
Pass B bundle	scripts/540_rho_ablation_bundle.py → alex_rho_ablation_v0/
Notebook	540_three_step_sim.ipynb
Binding	3-Master_Plan/BINDING_Selection_is_its_own_step.md
Slide	3-Master_Plan/re_entry/Model.pdf

Export naming

sports/datasets/mbb/exports_inverted_u_v0/
alex_side_by_side_v0/ # Pass A (λ knockout)
alex_rho_ablation_v0/ # Pass B (ρ ablation)

Pass B artifacts: generative_rho_low_16quantile.csv , generative_rho_high_16quantile.csv , generative_sort_chop_16quantile.csv , rho_ablation_summary.txt , rho_ablation_caption.txt , PNG.

Anti-patterns

- Opening archive/ notebooks as the daily workspace
- Merging L_net and S_i in comments or claims
- Saying ρ ablation **proves** hero bin-for-bin match
- Using τ in new user-facing text (archive reference only)

When stuck

Question	Read
What are we proving?	re_entry/02 , Model.pdf
Assignment math (legacy τ)	documents/Tier1_Presorting_Design_Note.md
Widget-era manual	documents/538_Cell10_Generative_Manual.md (archive context)
Implementation	tier1_pool_assignment.py — soft_assign , simulate_generative_rosters